

Proof. **1. Norm via the inner product.** By definition

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad \|y\| = \sqrt{\langle y, y \rangle}.$$

2. Degenerate cases. If $x = 0$ then $\|x + y\| = \|y\| = \|x\| + \|y\|$; similarly for $y = 0$. Hence we may assume $x \neq 0$ and $y \neq 0$.

3. Squaring the inequality. Since both sides are non-negative,

$$\|x + y\| \leq \|x\| + \|y\| \iff \|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

Using $(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2$ and $\|z\|^2 = \langle z, z \rangle$, inequality (1) becomes

$$\langle x + y, x + y \rangle \leq \langle x, x \rangle + 2\|x\|\|y\| + \langle y, y \rangle. \quad (2)$$

4. Expand the left-hand side. Bilinearity and $\langle u, v \rangle = \overline{\langle v, u \rangle}$ give

$$\langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \quad (1)$$

$$= \langle x, x \rangle + 2\operatorname{Re}\langle x, y \rangle + \langle y, y \rangle. \quad (3)$$

5. Apply Cauchy–Schwarz. The Cauchy–Schwarz inequality yields

$$|\langle x, y \rangle| \leq \|x\| \|y\|,$$

hence

$$2\operatorname{Re}\langle x, y \rangle \leq 2|\langle x, y \rangle| \leq 2\|x\| \|y\|. \quad (4)$$

6. Combine (3) and (4). Substituting the bound (4) into (3) we obtain exactly (2).

7. Take square roots. Both sides of (2) are non-negative, so

$$\|x + y\| = \sqrt{\langle x + y, x + y \rangle} \leq \sqrt{\|x\|^2 + 2\|x\|\|y\| + \|y\|^2} = \|x\| + \|y\|,$$

which is the desired triangle inequality.

8. Equality condition. Equality can occur only where equality holds in (4), i.e. in the Cauchy–Schwarz step. Equality in Cauchy–Schwarz holds iff $y = \lambda x$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = \lambda$; the requirement that $2\operatorname{Re}\langle x, y \rangle = 2\|x\| \|y\|$ forces λ to be a non-negative real number. Together with the degenerate cases $x = 0$ or $y = 0$ we obtain

$$\|x + y\| = \|x\| + \|y\| \iff (x = 0) \vee (y = 0) \vee (y = \lambda x \text{ with } \lambda \geq 0).$$

Thus the triangle inequality holds in every inner product space, and equality occurs precisely when the two vectors are non-negative scalar multiples of each other (including the trivial zero-vector cases). $\square$